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Outer space waste compaction

Abstract

A waste compaction apparatus may include a housing and a connector arm. The housing may define a drive chamber and a compaction chamber and the connector arm may include a drive piston portion movably positioned within the drive chamber and a compaction piston portion movably positioned within the compaction chamber. The drive chamber may be selectively fluidly coupleable to the near-vacuum of outer space, such that the compaction piston portion of the connector arm is configured to move within the compaction chamber to compact waste material contained within the compaction chamber in response to movement of the drive piston portion of the connector arm within the drive chamber.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4086378	12/1977	Kam et al.	N/A	N/A
4273037	12/1980	Ruebesam	N/A	N/A
4444099	12/1983	Paleschuck	N/A	N/A
4917712	12/1989	Crigler	N/A	N/A
5005457	12/1990	Thornton, Jr. et al.	N/A	N/A
5104810	12/1991	Birbara et al.	N/A	N/A
6106270	12/1999	Hardin	N/A	N/A
7481159	12/2008	Poss	100/48	B30B 9/3082
9486774	12/2015	Verma et al.	N/A	N/A
2002/0185016	12/2001	Hoffjann et al.	N/A	N/A
2008/0098909	12/2007	Willamor	N/A	N/A
2011/0259210	12/2010	Reed et al.	N/A	N/A
2016/0115405	12/2015	Zubrin et al.	N/A	N/A
2016/0153351	12/2015	Turner	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106365401	12/2016	CN	N/A
4320860	12/1993	DE	N/A
920874	12/1946	FR	N/A
H01153398	12/1988	JP	N/A
H10175098	12/1997	JP	N/A

OTHER PUBLICATIONS

European Patent Office, European Search Report dated Nov. 30, 2018 in Application No. 18176084.4. cited by applicant

USPTO; Requirement for Restriction dated Jan. 3, 2020 in U.S. Appl. No. 15/618,581. cited by applicant

USPTO; Non-Final Office Action dated Jun. 26, 2020 in U.S. Appl. No. 15/618,581. cited by

applicant

NASA, The Space Station Integrated Refuse Management System, 1998, p. xi, 8, 158-161 (Year: 1988). cited by applicant

USPTO; Final Office Action dated Dec. 15, 2020 in U.S. Appl. No. 15/618,581. cited by applicant

USPTO; Advisory Action dated Jan. 27, 2021 in U.S. Appl. No. 15/618,581. cited by applicant

USPTO; Non-Final Office Action dated Apr. 21, 2021 in U.S. Appl. No. 15/618,581. cited by applicant

USPTO; Final Office Action dated Oct. 21, 2021 in U.S. Appl. No. 15/618,581. cited by applicant

USPTO; Non-Final Office Action dated Mar. 18, 2022 in U.S. Appl. No. 15/618,581. cited by applicant

Agi, How Does Pneumatic Air Cylinder Actuators Work?, Feb. 2, 2011, p. 1 (Year: 2011). cited by applicant

USPTO; Notice of Allowance dated Oct. 12, 2022 in U.S. Appl. No. 15/618,581. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of, claims priority to and the benefit of, U.S. patent application Ser. No. 15/618,581, filed Jun. 9, 2017, and titled "OUTER SPACE WASTE COMPACTION," which is incorporated by reference herein in its entirety for all purposes.

FIELD

(1) The present disclosure relates to compacting systems, and more specifically, to utilizing the near-vacuum of outer space to compact and process waste material.

BACKGROUND

(2) Motorized actuation systems and devices used aboard spacecraft are often the result of extensive, timely, and costly development. For example, developing a motorized actuation device for a specific application onboard a spacecraft, including its associated controller, involves a substantial time and capital investment. Additionally, such devices are often heavy, thus adding weight to the spacecraft.

(3) Additionally, conventional Waste Collector Systems (WCS) aboard, for example, the Space Shuttle and International Space Station (ISS), utilize air flow to entrain human waste. For example, fecal material is typically collected by air flow entrainment. For instance, air may enter in a fairly uniform radial direction under a seat of the WCS. Air drag then helps separate the fecal bolus and draw it into a tank or fecal collection bag. The collected feces are generally thrown overboard. Because a substantial portion of fecal matter is comprised of water, disposing of collected fecal matter in this manner results in a loss of water, a valuable resource for space travel, from the spacecraft.

SUMMARY

(4) In various embodiments, the present disclosure provides a waste compaction apparatus that includes a housing and a connector arm. The housing may define a drive chamber and a compaction chamber. The connector arm may include a drive piston portion movably positioned within the drive chamber and a compaction piston portion movably positioned within the compaction chamber. The drive chamber is selectively fluidly coupleable to the near-vacuum of outer space, according to various embodiments, and the compaction piston portion of the connector

arm is configured to move within the compaction chamber to compact waste material contained within the compaction chamber in response to movement of the drive piston portion of the connector arm within the drive chamber, according to various embodiments.

(5) In various embodiments, the compaction piston portion of the connector arm is configured to move within the compaction chamber in a first direction to compact the waste material contained within the compaction chamber in response to movement of the drive piston portion of the connector arm within the drive chamber in the first direction. In various embodiments, the drive chamber is an annular chamber that circumscribes the compaction chamber. In various embodiments, the waste compaction apparatus further includes a diverter valve coupled to the housing that is in fluidic communication with the drive chamber. The diverter valve is selectively operable to control a drive pressure in the drive chamber, according to various embodiments. In various embodiments, the diverter valve is selectively fluidly coupleable to a waste gas source.

(6) The waste compaction apparatus may further include a heater coupled to the housing and configured to heat the waste material contained within the compaction chamber. The housing may include a fluid outlet from the compaction chamber. The connector arm may have a rigid body extending between the compaction piston portion and the drive piston portion.

(7) Also disclosed herein, according to various embodiments, is a waste processing system that includes, a connector arm, a diverter valve, and a gas-liquid separator. The housing may define a drive chamber, a compaction chamber, and a fluid outlet from the compaction chamber. The connector arm may include a drive piston portion movably positioned within the drive chamber and a compaction piston portion movably positioned within the compaction chamber. The diverter valve may be coupled to the housing in fluidic communication with the drive chamber. The gas-liquid separator may be coupled to the housing in fluidic communication with the compaction chamber via the fluid outlet. In various embodiments, the drive chamber is selectively fluidly coupleable to the near-vacuum of outer space via the diverter valve and the compaction piston portion of the connector arm is configured to move within the compaction chamber to compact waste material contained within the compaction chamber in response to movement of the drive piston portion of the connector arm within the drive chamber. In various embodiments, in response to compacting the waste material contained within the compaction chamber, the compaction chamber is configured to expel a waste fluid via the fluid outlet to the gas-liquid separator. In various embodiments, the gas-liquid separator is configured to separate the waste fluid into waste liquid and waste gas.

(8) In various embodiments, the waste processing system further includes a first conduit extending between the diverter valve and a first portion of the drive chamber, and a second conduit extending between the diverter valve and a second portion of the drive chamber. The first portion of the drive chamber may be at least partially defined by a first side of the drive piston portion, and the second portion of the drive chamber is at least partially defined by a second side of the drive piston portion opposite the first side. The waste processing system may further include a waste gas conduit configured to direct the waste gas from the gas-liquid separator. The waste gas conduit may be configured to direct the waste gas to outer space or to a waste gas holding tank. The waste gas holding tank is fluidly coupled to the diverter valve to provide back-fill gas to the drive chamber, according to various embodiments. The waste gas valve is configured to selectively direct the waste gas to at least one of outer space and a waste gas holding tank. The waste processing system may further include a heater coupled to the housing that is configured to heat the waste material contained within the compaction chamber.

(9) Also disclosed herein, according to various embodiments, is a method for processing waste. The method includes exposing a drive chamber of a housing to the near-vacuum of outer space and compacting waste material within a compaction chamber of the housing. The waste material may be fecal matter, and the method may further include, in response to the compacting the fecal matter, separating waste fluid from the compaction chamber into a waste gas and a waste liquid via a gas-

liquid separator. Still further, the method may include collecting the waste liquid for recycling.

(10) The forgoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a cross-sectional view of a waste compaction apparatus in a pre-compacted state, in accordance with various embodiments;
 - (2) FIG. 2 illustrates a cross-sectional view of a waste compaction apparatus in a compacted state, in accordance with various embodiments;
 - (3) FIG. 3 illustrates a cross-sectional view of a waste processing system, with various components of the system schematically depicted, in accordance with various embodiments; and
 - (4) FIG. 4 is a schematic flowchart diagram of a method for processing waste, in accordance with various embodiments.
- (5) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the detailed description and claims when considered in connection with the drawing figures, wherein like numerals denote like elements.

DETAILED DESCRIPTION

(6) The detailed description of exemplary embodiments herein makes reference to the accompanying drawings, which show exemplary embodiments by way of illustration. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that logical changes and adaptations in design and construction may be made in accordance with this disclosure and the teachings herein without departing from the spirit and scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation.

(7) In various embodiments, and with reference to FIG. 1, a waste compaction apparatus **100** is provided. The waste compaction apparatus **100**, as described in greater detail below, generally takes advantage of the near-vacuum of outer space to compact waste material. In various embodiments, the waste compaction apparatus **100** may be used to compact fecal matter, trash, garbage, or other waste. In various embodiments, the waste compaction apparatus **100** is used to extract and recycle water from fecal matter. The waste compaction apparatus **100** may be utilized with a system, such as waste processing system **10** described below with reference to FIG. 3. While numerous details are included herein pertaining to using the waste compaction apparatus as a feces processing apparatus, the scope of the present disclosure is not limited to feces processing. The waste compaction apparatus **100** (with reference to FIGS. 1 and 2) and the waste processing system **10** (with reference to FIG. 3) may be implemented for use within the ISS, low earth orbit (LEO) aircraft, crew exploration vehicle (CEV), and/or a space hotel to compact and/or otherwise process waste material. Stated another way, the waste compaction apparatuses, systems, and methods disclosed herein may supplement, replace, and/or be utilized in conjunction with existing aerospace waste collector systems.

(8) The waste compaction apparatus **100**, according to various embodiments and with continued reference to FIG. 1, includes a housing **110** and a connector arm **120**. The housing **110** defines a drive chamber **112** and a compaction chamber **114** and the connector arm includes a drive piston portion **122** movably positioned within the drive chamber **112** and a compaction piston portion **124**

movably positioned within the compaction chamber **114**, according to various embodiments. Generally, waste material **60**, such as fecal matter, is placed into the compaction chamber **114** and the waste compaction apparatus **100** is configured to compact the waste material **60**. The compacting force, according to various embodiments, is provided by the near-vacuum of outer space, as described in greater detail below, as opposed to utilizing a motorized compaction device. In various embodiments, the compaction of the waste material **60** reduces the volume of the waste onboard a spacecraft. In various embodiments, as described in greater detail below with reference to FIG. 3, the waste compaction apparatus **100** may be utilized to extract waste liquid (e.g., water) from fecal matter for collection/recycling. FIG. 1, according to various embodiments, illustrates the waste compaction apparatus **100** in a pre-compacted state and FIG. 2, according to various embodiments, illustrates the waste compaction apparatus **100** in a compacted state.

(9) In various embodiments, and with reference to FIGS. 1 and 2, the drive piston portion **122** is configured to move within the drive chamber **112**, and the compaction piston portion **124** is configured to move within the compaction chamber **114**. For example, the drive chamber **112** may be selectively fluidly coupleable to the near-vacuum of outer space, such that, in response to exposing at least a portion of the drive chamber **112** to the near-vacuum of outer space, said portion of the drive chamber **112** may be substantially evacuated, thereby creating a pressure differential across the drive piston portion **122**, causing the drive piston portion **122** to move. For example, the pressure on one side of the drive piston portion **122** may be about 1 atmosphere (101.3 kilopascals) while the pressure on the other, evacuated side of the drive piston portion **122** may be less than about 1 micropascal. In response to this movement of the drive piston portion **122**, the compaction piston portion **124** of the connector arm **120** may also move within the compaction chamber **114** to compact the waste material **60** contained within the compaction chamber **114**. By compacting certain types of waste materials, such as fecal matter, waste fluid may be urged and extracted from the fecal matter in response to compaction and may exit through fluid outlet **117**. This waste fluid, as described in greater detail below, may be further processed and refined in order to extract waste liquid (e.g., water) from the waste material **60**.

(10) In various embodiments, the waste compaction apparatus **100** may further include a heater **105** coupled to the housing **110** and configured to provide heat to the waste material **60**. The heater **105** may be configured to elevate the temperature of the waste material **60**, thereby facilitating extraction of waste fluid from the waste material **60** (e.g., water from fecal matter) and also helping to kill bacteria or other microbes/microorganisms present in the waste material **60**. According to various embodiments, the heater **105** may raise the temperature of the waste material **60** to between about 50 degrees Celsius and about 250 degrees Celsius. In various embodiments, the heater **105** may raise the temperature of the waste material **60** to between about 100 degrees Celsius and about 200 degrees Celsius. As used in this context only, the term “about” refers to plus or minus 5% of the indicated value. Accordingly, the waste compaction apparatus **100** may be a torrefaction-type process that includes heating and compaction functionality.

(11) In various embodiments, the heater **105** may be coupled to and/or disposed around side walls **113** of the portion of the housing **110** that defines the compaction chamber **114**. In various embodiments, the housing **110** may include other walls, such as a moveable piston-like wall **111** and a base wall **115**, that collectively define the compaction chamber **114**. The moveable piston-like wall **111** may be coupled to, and/or may be in abutting engagement with, the compaction piston portion **124** of the connector arm **120**. The housing **110** may also include other walls that define the drive chamber **112**. In various embodiments, the housing **110** may be comprised of multiple portions or sections that respectively form and define the drive chamber **112** and the compaction chamber **114**. For example, the housing **110** may comprise two separate housings, a first housing defining the drive chamber **112** and a second housing defining the compaction chamber **114**, and the two housings may be rigidly fixed relative to each other (i.e., preventing relative movement between the two housings). However, in various embodiments the housing **110**

may be integrally formed of a single piece of material.

(12) In various embodiments, the compaction chamber **114** of the housing **110** is a central chamber, and the drive chamber **112** is an annular chamber formed around the compaction chamber **114** (i.e., the drive chamber **112** circumscribes the compaction chamber **114**). Accordingly, the connector arm **120** may have an annular configuration, with the compaction piston portion **124** having a central plate configured to move within the compaction chamber **114**, and the drive piston portion **122** having an annular ring configured to move within the annulus that is the drive chamber **112**. In various embodiments, the housing **110** is cylindrical, and thus the defined chambers **112**, **114** therein and the piston portions **122**, **124** of the connector arm **120** may also be cylindrical.

(13) In various embodiments, the connector arm **120** has a rigid body that extends between the drive piston portion **122** and the compaction piston portion **124**. For example, the connector arm **120** may be integrally formed of a single, unitary material. The piston portions **122**, **124** of the connector arm **120**, according to various embodiments, are the fluid/waste facing surfaces that directly face and engage the respective contents of the chambers **112**, **114**. Thus, the piston portions **122**, **124** may include sealing features that facilitate a fluid seal between the piston portions **122**, **124** and the respective walls of the housing **110**. In various embodiments, the piston portions **122**, **124** engage other components, such as the moveable piston-like wall **111** of the housing **110**, and said other components form a fluid seal with adjacent and respective walls of the housing **110**. In various embodiments, an annular seal **109** may be utilized in conjunction with the moveable piston-like wall **111** and the compaction piston portion **124** of the connector arm **120**. Said differently, the waste compaction apparatus **100** may include an annular seal **109** that is disposed radially outward of the compaction piston portion **124** of the connector arm **120**, and the annular seal **109** may engage a radially inward surface of side walls **113** of the housing **110**. The connector arm **120** may include a segment/section that directly couples to and/or abuts the annular seal **109**.

(14) In various embodiments, movement of the piston portions **122**, **124** of the connector arm **120** are in the same direction. Said differently, the compaction piston portion **124** of the connector arm **120** may be configured to move within the compaction chamber **114** in a first direction to compact the waste material **60** in response to the driven movement (suction of the near-vacuum of outer space) of the drive piston portion **122** of the connector arm **120** within the drive chamber **112** in the first (e.g., same) direction. Similarly, the compaction piston portion **124** of the connector arm **120** may be configured to move within the compaction chamber **114** in a second direction opposite the first direction to retract the compaction piston portion **124** in response to retraction of the drive piston portion **122** of the connector arm **120** within the drive chamber **112** in the second direction.

(15) As mentioned above, the process of compacting (and optionally heating) the waste material **60** facilitates the extraction of waste fluid from the waste matter **60** (e.g., water from fecal matter). The waste fluid may seep out of the waste material due, at least in part, to the compression forces exerted on the waste material **60** by the compaction piston portion **124** of the connector arm **120** and/or the waste fluid may boil out of the waste material due to the heating of the waste material **60**. The waste compaction apparatus **100** may also include a solid/fluid separator **119** disposed in the fluid outlet **117** from the compaction chamber **114** that helps to retain the solid waste material **60** within the compaction chamber **114** during compaction. The extracted waste fluid, as described in greater detail below with reference to FIG. 3, may be further processed and refined in order to recycle/reuse at least the extracted water content.

(16) The waste compaction apparatus **100** may further include a diverter valve **130** that is coupled to the housing **110**. The diverter valve **130** may be in fluidic communication with the drive chamber **112**. The diverter valve **130** may be selectively operable to control a drive pressure in the drive chamber **112**. The drive pressure may be defined as the pressure differential across the drive piston portion **122**. In various embodiments, the diverter valve **130** is fluidly coupled to outer space **50** (or any other near-vacuum) and may be selectively controlled to expose the drive chamber **112** to the near-vacuum pressure of outer space **50** in order to drive motion of the drive piston portion

122, which drives compaction of the waste material **60** via force transferred through the connector arm **120** to the compaction piston portion **124**. The diverter valve **130** may also be selectively coupleable to a waste gas source, as described in greater detail below with reference to FIG. 3. (17) In various embodiments, and with continued reference to FIGS. 1 and 2, the waste compaction apparatus **100** may include a first conduit **136** and a second conduit **138**. The first conduit **136** may extend between the diverter valve **130** and a first portion **116** of the drive chamber **112** and the second conduit **138** may extend between the diverter valve **130** and a second portion **118** of the drive chamber **112**. The first and second portions **116**, **118** of the drive chamber **112** may be disposed on opposite sides of the moveable drive piston portion **122** of the connector arm **120**. Said differently, the first portion **116** of the drive chamber **112** is at least partially defined by a first side **126** of the drive piston portion **122** and the second portion **118** of the drive chamber **112** is at least partially defined by a second side **128** of the drive piston portion **122**, wherein the first side **126** of the drive piston portion **122** is opposite the second side **128**.

(18) The diverter valve **130** may be selectively controlled and actuated to, for example, fluidly connect the near-vacuum of outer space **50** to the second conduit **138**, thereby evacuating the second portion **118** of the drive chamber **112** and creating the aforementioned pressure differential across the drive piston portion **122**, which causes the movement of the connector arm **120**. The diverter valve **130** may be configured to direct back-flow of a gas, for example, a waste gas from a waste gas source, to the first portion **116** of the drive chamber **112** via the first conduit **136** during compaction. After a completed compaction step, the diverter valve **130** may be configured introduce a waste gas into the second portion **118** of the drive chamber **112** via the second conduit **138** to increase the pressure in the second portion **118** of the drive chamber **112**, thereby decreasing (or eliminating or reversing) the pressure differential across the drive piston portion **122** to allow the compaction piston portion **124** of the connector arm **120** to be retracted from the compaction chamber **114**. The spent/compacted waste material **60** may be disposed of or may be utilized for radiation shielding, among other uses, and a new batch of waste material **60** may be loaded into the compaction chamber **114**.

(19) In various embodiments, and with reference to FIG. 3, a waste processing system **10** is provided. As mentioned above, the waste processing system **10** may include, in addition to the components of the waste compaction apparatus **100** described above, one or more additional refining/processing units that facilitate water recycling from the waste material **60**. In various embodiments, the waste processing system **10** includes the housing **110**, the connector arm **120**, and the diverter valve **130** as described with reference to FIGS. 1 and 2. The waste processing system **10** may further comprise a gas-liquid separator **140** coupled to the housing **110**. The gas-liquid separator **140** may be in fluidic communication with the compaction chamber **114** via one or more fluid outlets **117**. One or more valves or filters **141** may be disposed along the conduit between the compaction chamber **114** and the gas-liquid separator **140**. The gas-liquid separator **140** may include filters or membranes **143** that are configured to separate the waste fluid from the compaction chamber **114** into a waste liquid (e.g., water) and a waste gas. The waste liquid may be further refined, deodorized, etc., in order to render the waste liquid useable.

(20) The waste processing system **10** may further include a waste gas conduit **142** in fluid communication with the gas-liquid separator **140** and a waste gas valve **145**. The waste gas conduit **142** may be configured to direct the waste gas from the gas-liquid separator **140** to be dumped into outer space **50** or to be directed to a waste gas holding tank **150** (e.g. a waste gas source). For example, the waste processing system **10** may include the waste gas valve **145** that allows the waste gas to be selectively directed to either vent to outer space **50** via conduit **146** and/or flow to the waste gas holding tank via conduit **147**. The waste gas holding tank **150** may be fluidly coupled to the diverter valve **130** and, as described above, the diverter valve **130** may be selectively controlled to utilize the waste gas to back-fill portions of the drive chamber **112** and/or to return the drive chamber **112** to a nominal pressure after compaction. In various embodiments, the waste

processing system **10** may further include one or more heating units **148** that are configured to introduce heat into the system in order to prevent the low temperature of the near-vacuum environment of outer space from adversely affecting the components of the system **10**.

(21) Also disclosed herein, according to various embodiments and with reference to FIG. **4**, is a method **490** for processing waste. The method **490** may include exposing a drive chamber to the near-vacuum of outer space at step **492**. The method **490** may further include separating waste fluid from waste material at step **494** and collecting waste liquid from the waste fluid at step **496**. Once again, although numerous details and examples are included herein pertaining to the compaction of feces, those skilled in the art will recognize that the systems and methods disclosed herein may be utilized with other waste matter, such as trash or garbage.

(22) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure.

(23) The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” It is to be understood that unless specifically stated otherwise, references to “a,” “an,” and/or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. All ranges and ratio limits disclosed herein may be combined.

(24) Moreover, where a phrase similar to “at least one of A, B, and C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(25) The steps recited in any of the method or process descriptions may be executed in any order and are not necessarily limited to the order presented. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Elements and steps in the figures are illustrated for simplicity and clarity and have not necessarily been rendered according to any particular sequence. For example, steps that may be performed concurrently or in different order are illustrated in the figures to help to improve understanding of embodiments of the present disclosure.

(26) Any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. Surface shading lines may be used throughout the figures to denote different parts or areas but not necessarily to denote the same or different materials. In some cases, reference coordinates may be specific to each figure.

(27) Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or

characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(28) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element is intended to invoke 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

Claims

1. A waste compaction apparatus comprising: a housing defining a drive chamber, a compaction chamber, and a fluid outlet from the compaction chamber; and a connector arm comprising a drive piston portion movably positioned within the drive chamber and a compaction piston portion movably positioned within the compaction chamber; a diverter valve coupled to the housing in fluidic communication with the drive chamber; a first conduit extending between the diverter valve and a first portion of the drive chamber, wherein the first portion of the drive chamber is at least partially defined by a first side of the drive piston portion; a second conduit extending between the diverter valve and a second portion of the drive chamber, the second portion of the drive chamber being defined, at least, partially a second side of the drive piston portion opposite the first side of the drive piston portion; a waste gas source fluidly coupled to the diverter valve, and a gas-liquid separator coupled to the housing in fluidic communication with the compaction chamber via the fluid outlet; wherein; the second portion of the drive chamber is selectively fluidly coupleable to a near-vacuum of outer space via the diverter valve, wherein, during compaction, the diverter valve fluidly couples the second portion of the drive chamber to the near-vacuum of outer space, and wherein, after compaction, the diverter valve directs gas from the waste gas source into the second portion of the drive chamber via the second conduit to retract the compaction piston portion of the connector arm; the compaction piston portion of the connector arm is configured to move within the compaction chamber to compact waste material contained within the compaction chamber in response to movement of the drive piston portion of the connector arm within the drive chamber; the connector arm comprises an annular configuration, with the compaction piston portion having a central plate; in response to compacting the waste material contained within the compaction chamber, the compaction chamber is configured to expel a waste fluid via the fluid outlet to the gas-liquid separator; and the gas-liquid separator is configured to separate the waste fluid into waste liquid and waste gas.

2. The waste compaction apparatus of claim 1, wherein the compaction piston portion of the connector arm is configured to move within the compaction chamber in a first direction to compact the waste material contained within the compaction chamber in response to movement of the drive piston portion of the connector arm within the drive chamber in the first direction.

3. The waste compaction apparatus of claim 2, wherein the drive chamber is an annular chamber that circumscribes the compaction chamber.

4. The waste compaction apparatus of claim 1, wherein the diverter valve is selectively operable to control a drive pressure in the drive chamber.

5. The waste compaction apparatus of claim 1, further comprising a heater coupled to the housing and configured to heat the waste material contained with the compaction chamber.

6. The waste compaction apparatus of claim 1, wherein the connector arm has a rigid body extending between the compaction piston portion and the drive piston portion.
